Huynh Quoc Viet

AI Researcher & Machine Learning Engineer

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SUMMARY

Artificial intelligence engineering student and research team lead working across trustworthy machine learning, conformal prediction, multimodal anomaly detection, time series forecasting, computer vision, and reinforcement learning. Research includes an accepted AJCAI 2026 poster, supported by hands-on implementation with PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, FastAPI, and Streamlit.

EDUCATION

FPT University Ho Chi Minh City, Vietnam
Bachelor of Engineering in Artificial Intelligence *Expected Jan 2027* | *GPA: 3.12/4.0*

- **Honors:** Honorable Student of the Semester (FALL2025).

HELP University Kuala Lumpur, Malaysia
Certification: English Proficiency Program *Feb 2024*

TECHNICAL SKILLS

Programming: Python, SQL, Java (basic), JavaScript/TypeScript, LaTeX.
Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, Hugging Face, OpenCV, YOLO, RT-DETR.
Methods: Conformal Prediction, Multimodal Anomaly Detection, Computer Vision, Time-Series Forecasting, Reinforcement Learning, Explainable AI.
Data & Engineering: Pandas, NumPy, FastAPI, Streamlit, PostgreSQL, Supabase, Docker, GitHub Actions, Linux, Weights & Biases.

SELECTED PUBLICATIONS

Warning Before the Evening Peak: Health-Centred Air Quality Alerts in New Zealand and Australia
AJCAI 2026 **CORE B** | *Accepted Poster*

- Built calibrated 1–3 day air-quality alerts for New Zealand and Australia using weather forecasts, atmospheric inversion features, and machine learning models focused on health-relevant evening peaks.

Trustworthy Multimodal Anomaly Detection in Time Series with Conformal Calibration and Counterfactual Attribution
KDD 2027 — Research Track, Cycle 1 **CORE A*** | *Decision pending*

- Built the **C-DLE** framework to combine numeric and text signals, control false alarms with split-conformal calibration, and explain which modality triggered each alert.

Distribution-Free Coverage of Attack Classes on Devices Absent from Training
AISTATS 2027 **CORE A** | *Under Review*

- Developed hierarchical class-conditional conformal prediction to quantify attack-class uncertainty on unseen IoT devices without assuming packet-level exchangeability.

Stress-Testing Multi-Source Cyanobacterial Bloom Forecasting under Sparse Monitoring
AJCAI 2026 **CORE B** | *Decision pending*

- Combined water chemistry, weather, and Sentinel-2 data to forecast cyanobacterial blooms across seven lakes, with robustness tests for sparse and missing observations.

RESEARCH EXPERIENCE

SpeedyLabX — Student Research Team FPT University
AI Research Team Member (Tech Lead) *Jan 2025 – Present*

- Co-authored two conference submissions on distribution-free IoT attack classification and multi-source cyanobacterial bloom forecasting.
- Lead an ongoing regime-adaptive stock forecasting study using **HMM-LSTM**, OHLCV features, **SHAP**, and walk-forward evaluation.

PROJECTS

FitSync — Personal Training Workspace EXE201 Startup Project
Project Lead — Product & OCR *Sep 2026 – Nov 2026*

- Coordinate a five-member team building a mobile-first coaching workspace for Vietnamese personal trainers with **Next.js**, Supabase/PostgreSQL, and row-level security.
- Own product scope and delivery evidence; co-lead privacy-aware InBody OCR evaluation with mandatory coach review and manual fallback.

Adaptive Preference Allocation for Multi-Objective RL REL301m Course Project
Research Engineer *Sep 2026 – Nov 2026*

- Implemented a preference-conditioned vector Q-network with per-head **Double-DQN** targets and adaptive allocation using performance gap, uncertainty, coverage, and learning-progress signals.
- Built compute-matched controls, paired-seed experiment orchestration, and reproducible evaluation; multi-seed course experiments are in progress.

Personal Protective Equipment (PPE) Detection DPL302m Course Project
Deep Learning Engineer *Sep 2024 – Dec 2024*

- Unified nine public datasets and benchmarked **YOLOv8** and RT-DETR models for accuracy–latency trade-offs.
- Deployed real-time webcam inference and automated violation reporting through a **Streamlit** application.

Face Recognition Attendance System CPV301 Course Project
Model Engineer *Jun 2024 – Aug 2024*

- Built an **OpenCV**/Tkinter attendance application using LBPH recognition and Haar Cascade detection, with support for lighting variation and partial occlusion.

COMPETITIONS

NASA Space Apps Challenge Ho Chi Minh City
Data & Model Engineer — CosmicOptic *2025*

- Built a **Kepler** light-curve pipeline for transit detection and orbital-feature extraction, integrated with a **FastAPI** visualization application.

FPT Hackathon FPT University
Competitive Programming Participant *May 2025*

- Applied dynamic programming and graph algorithms in an ICPC-style contest under strict time limits.

CERTIFICATIONS

- **IBM:** AI Foundations for Everyone; Data Science Fundamentals with Python and SQL; Full Stack Software Developer.
- **NVIDIA:** Building RAG Agents with LLMs; Developing an AI Background Generator with NIM.